

6.5 均匀平面波垂直投射到理想导电平面上

已知 $\vec{E}^+ = \hat{x}E_{x0}^+ e^{-jkz}$

$$\vec{E}^- = \hat{x}E_{x0}^- e^{jkz} \quad E_{x0}^- = ?$$

边界条件 $E_t = 0$

$$E_t(z=0) = 0 \quad E_{x0}^- = -E_{x0}^+$$

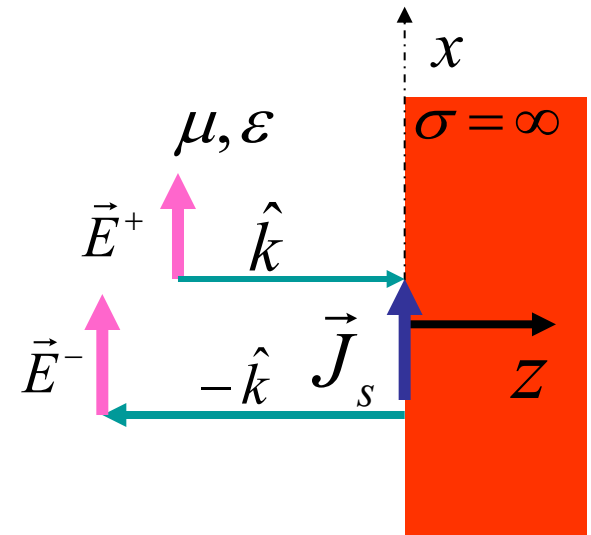
总电场 $\vec{E} = \vec{E}^+ + \vec{E}^-$

$$= E_{x0}^+ e^{-jkz} + E_{x0}^- e^{jkz}$$

$$= E_{x0}^+ (e^{-jkz} - e^{jkz})$$

$$= -j2E_{x0}^+ \sin(kz)$$

$$E_x(t) = E(t) = 2\sqrt{2}E_{x0}^+ \sin(kz) \sin(\omega t) \quad \text{驻波}$$



$$E_x = E = -j2E_{x0}^+ \sin(kz)$$

$$E_x(t) = 2\sqrt{2}E_{x0}^+ \sin(kz) \sin(\omega t)$$

电场波节点 $kz = -n\pi \quad z_{\min} = -n\frac{\lambda}{2}$

电场波腹点 $kz = -n\pi - \frac{\pi}{2} \quad z_{\max} = -n\frac{\lambda}{2} - \frac{\lambda}{4}$

磁场 $\vec{H}^+ = \frac{\hat{z} \times \vec{E}^+}{Z} = \hat{y} \frac{E_{x0}^+}{Z} e^{-jkz}$

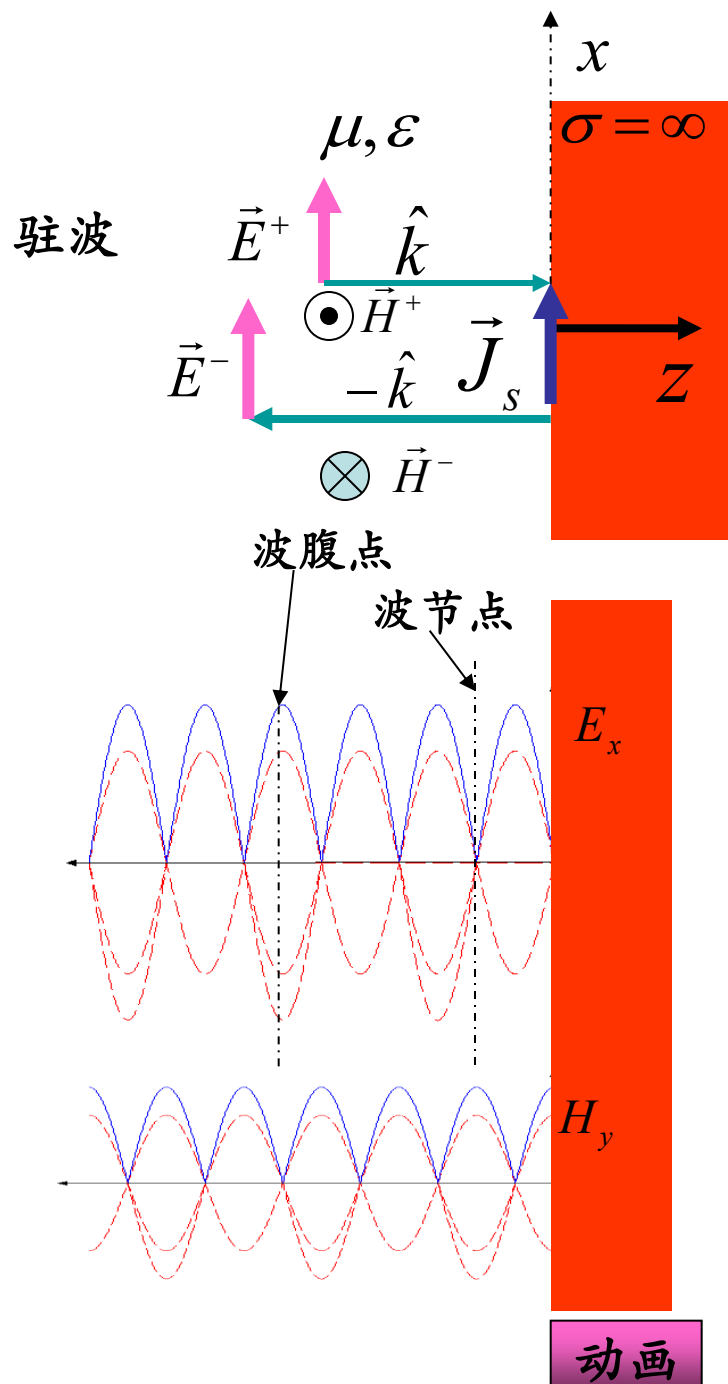
$$\vec{H}^- = \frac{-\hat{z} \times \vec{E}^-}{Z} = -\hat{y} \frac{E_{x0}^-}{Z} e^{jkz} = \hat{y} \frac{E_{x0}^+}{Z} e^{jkz}$$

$$H_y = H^+ + H^- = 2 \frac{E_{x0}^+}{Z} \cos(kz)$$

$$H_y(t) = 2\sqrt{2} \frac{E_{x0}^+}{Z} \cos(kz) \cos(\omega t)$$

$$\vec{S}_c$$

$$\vec{J}_s$$

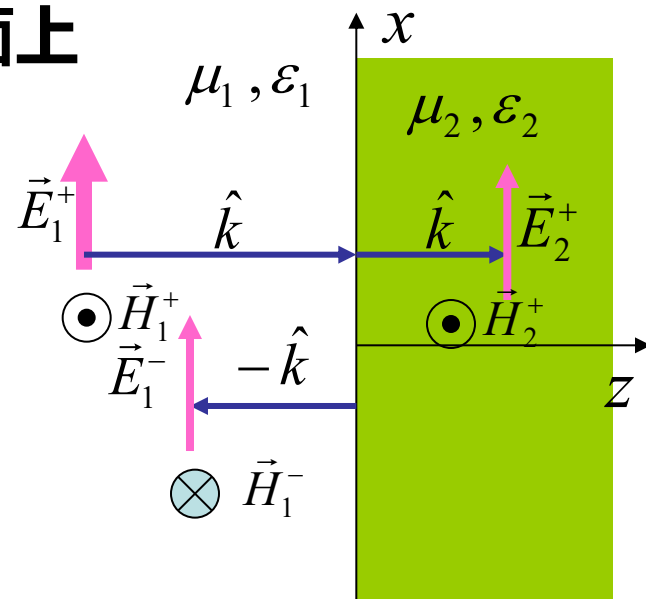


6.6 均匀平面波垂直投射到两种介质分界面上

已知

$$\begin{aligned}\vec{E}_1^+ &= \hat{x} E_{x10}^+ e^{-jk_1 z} & \vec{H}_1^+ &= \hat{y} \frac{E_{x10}^+}{Z_1} e^{-jk_1 z} \\ \vec{E}_1^- &= \hat{x} E_{x10}^- e^{jk_1 z} & \vec{H}_1^- &= -\hat{y} \frac{E_{x10}^-}{Z_1} e^{jk_1 z} \\ \vec{E}_2^+ &= \hat{x} E_{x20}^+ e^{-jk_2 z} & \vec{H}_2^+ &= \hat{y} \frac{E_{x20}^+}{Z_2} e^{-jk_2 z}\end{aligned}$$

$$k_2 = \omega \sqrt{\mu_2 \varepsilon_2} = \frac{2\pi}{\lambda_2}$$



界面反射系数

$$R = \frac{E_1^-(z=0)}{E_1^+(z=0)} = \frac{E_{x10}^-}{E_{x10}^+} \quad E_{x10}^- = RE_{x10}^+$$

界面传输系数

$$T = \frac{E_2^+(z=0)}{E_1^+(z=0)} = \frac{E_{x20}^+}{E_{x10}^+} \quad E_{x20}^+ = TE_{x10}^+$$

由 $z=0$ 边界条件

$$E_{1t} = E_{2t} \quad E_{x10}^+ + RE_{x10}^+ = TE_{x10}^+ \quad 1 + R = T$$

$$H_{1t} = H_{2t} \quad \frac{E_{x10}^+}{Z_1} - \frac{RE_{x10}^+}{Z_1} = \frac{TE_{x10}^+}{Z_2} \quad (1 - R)/Z_1 = T/Z_2$$

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

$$T = \frac{2Z_2}{Z_2 + Z_1}$$

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$$

$$T = \frac{2Z_2}{Z_2 + Z_1}$$

Z_1 Z_2 实数
 R T 实数

$Z_2 > Z_1$ $0 < R < 1, T > 1$
 $Z_2 < Z_1$ $-1 < R < 0, T < 1$

复数?

$$\vec{E}_1 = \vec{E}_1^+ + \vec{E}_1^-$$

$$E_1 = E_{x10}^+ (e^{-jk_1 z} + R e^{jk_1 z})$$

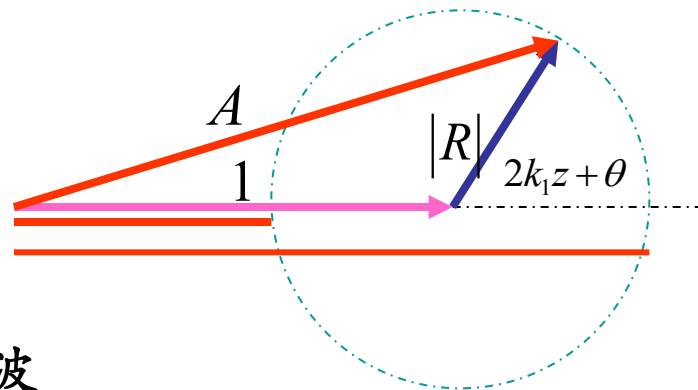
$$= E_{x10}^+ (e^{-jk_1 z} - R e^{-jk_1 z} + R e^{-jk_1 z} + R e^{jk_1 z})$$

$$= E_{x10}^+ \{ (1-R)e^{-jk_1 z} + 2R \cos(k_1 z) \}$$
 行波+驻波

$$= E_{x10}^+ (1 + R e^{j2k_1 z}) e^{-jk_1 z} \quad R = |R| e^{j\theta}$$

$$= E_{x10}^+ (1 + |R| e^{j(2k_1 z + \theta)}) e^{-jk_1 z}$$

$A(z)$

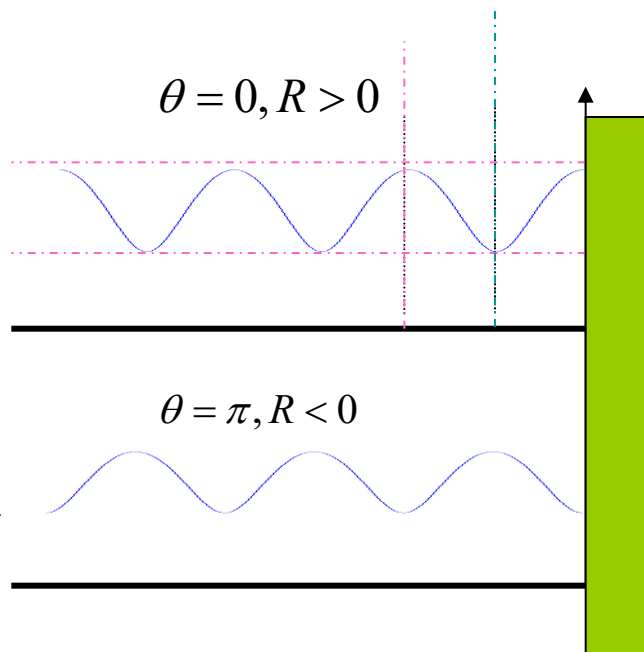


振幅随位置变化的行波

波腹点 $2k_1 z + \theta = \pm 2n\pi$ $z_{\max} = -\frac{\theta}{4\pi} \lambda_1 \pm n \frac{\lambda_1}{2}$

波节点 $2k_1 z + \theta = \pm (2n+1)\pi$ $z_{\min} = -\frac{\theta}{4\pi} \lambda_1 \pm (2n+1) \frac{\lambda_1}{4}$

电场驻波比 $\rho = \frac{E_{\max}}{E_{\min}} = \frac{E_{x10}^+ (1 + |R|)}{E_{x10}^+ (1 - |R|)}$ $\rho = \frac{1 + |R|}{1 - |R|}$



动画

△

- 输入阻抗

- 定义距分界面距离 l 的点上的 E 和 H 之比为该点的输入阻抗。

$$Z_{in}(z = -l) = E(z = -l) / H(z = -l)$$

波阻抗 $Z = E_{i,r,t} / H_{i,r,t} = \sqrt{\mu / \varepsilon}$ $R = \frac{Z_2 - Z_1}{Z_2 + Z_1}$

$$\bar{E}_1 = \hat{x} E_{x0}^+ (e^{-jk_1 z} + \text{Re} e^{jk_1 z}) \quad \bar{H}_1 = \hat{y} \frac{E_{x0}^+}{Z_1} (e^{-jk_1 z} - \text{Re} e^{jk_1 z})$$

$$Z_{in}(l) = \frac{E_1(z = -l)}{H_1(z = -l)} = \frac{e^{jk_1 l} + \text{Re} e^{-jk_1 l}}{e^{jk_1 l} - \text{Re} e^{-jk_1 l}} Z_1 = \frac{e^{jk_1 l} + \frac{Z_2 - Z_1}{Z_2 + Z_1} e^{-jk_1 l}}{e^{jk_1 l} - \frac{Z_2 - Z_1}{Z_2 + Z_1} e^{-jk_1 l}} Z_1$$

$$= \frac{(Z_2 + Z_1)e^{jk_1 l} + (Z_2 - Z_1)e^{-jk_1 l}}{(Z_2 + Z_1)e^{jk_1 l} - (Z_2 - Z_1)e^{-jk_1 l}} Z_1 = \frac{Z_2 \cos(k_1 l) + jZ_1 \sin(k_1 l)}{Z_1 \cos(k_1 l) + jZ_2 \sin(k_1 l)} Z_1$$

$$= \frac{Z_2 + jZ_1 \text{tg} k_1 l}{Z_1 + jZ_2 \text{tg} k_1 l} Z_1$$

例：均匀平面波从空气中垂直投射到导电媒质界面上. 在空气中测得距离界面最近的相邻两波腹波节点 $E_{\max}(l_{\max} = 7.5\text{cm}) = 5\text{V/m}$ $E_{\min}(l_{\min} = 20\text{cm}) = 1\text{V/m}$

求： f, R, Z_2

解： $\frac{\lambda}{4} = l_{\min} - l_{\max} = 12.5\text{cm}$ $\lambda = 50\text{cm}$ $f = \frac{c}{\lambda} = 600\text{MHz}$

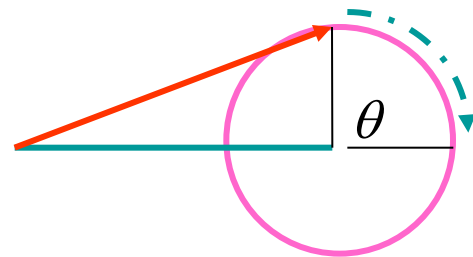
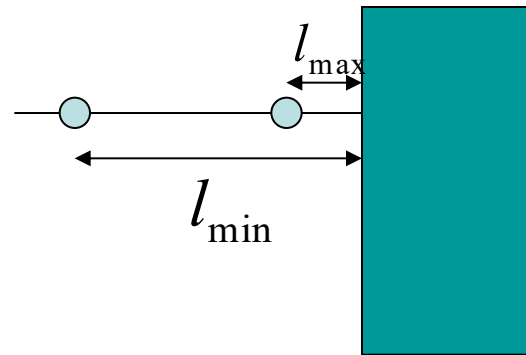
$$\rho = \frac{1 + |R|}{1 - |R|} \quad |R| = \frac{\rho - 1}{\rho + 1} = \frac{2}{3} \quad \rho = \frac{E_{\max}}{E_{\min}} = 5$$

$$R = |R|e^{j\theta} \quad 2kz + \theta = \pm 2n\pi$$

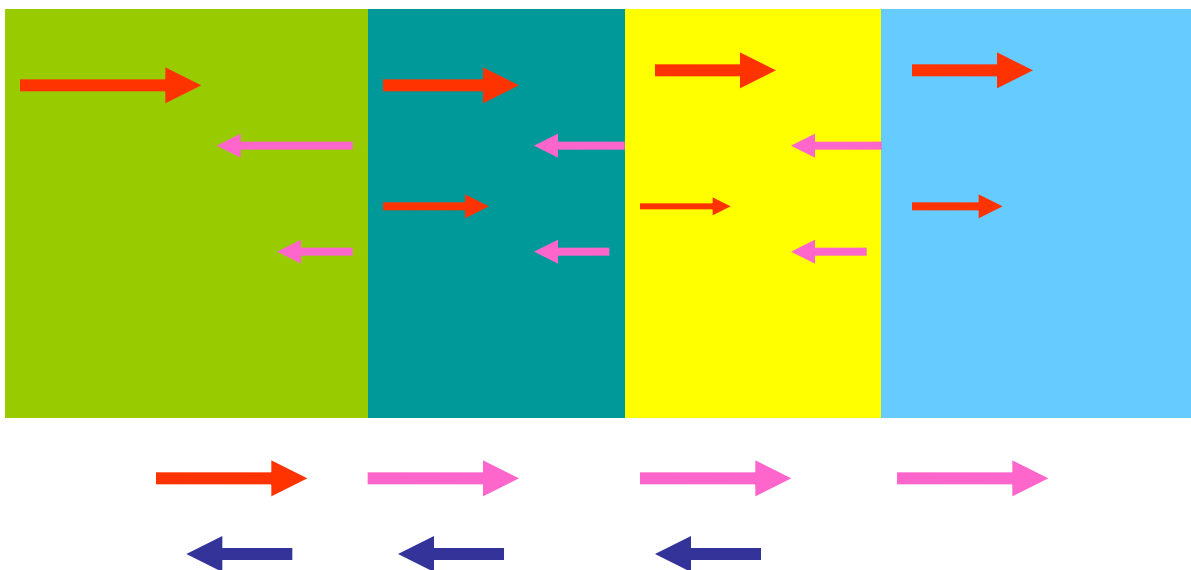
$$\theta = -2kz = 2kl_{\max} = \frac{4\pi}{\lambda}l_{\max} = 0.6\pi$$

$$R = \frac{2}{3}e^{j0.6\pi}$$

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} \quad Z_2 = \frac{1 + R}{1 - R}Z_1 = 90\pi e^{j67^\circ}$$



6.7 均匀平面波垂直入射到多层介质中



一般分析方法:

- (1) 写出已知的入射波
- (2) 根据已知入射波的性质,写出反射波以及其它各层中的正向波和反向波的一般表达式,共有 $2(n-1)$ 个未知量
- (3) 写出 $n-1$ 个边界上的电场和磁场的共 $2(n-1)$ 个边界条件方程
- (4) 解边界条件方程,得到 $2(n-1)$ 个未知的各区电磁波复振幅.

举例分析3层介质

建立坐标系

1) 已知入射波, 写各区的电磁波

$$E_{x1}^+ = E_{x10}^+ e^{-jk_1 z} \quad H_{y1}^+ = \frac{E_{x10}^+}{Z_1} e^{-jk_1 z}$$

$$E_{x1}^- = \underline{E_{x10}^-} e^{jk_1 z} \quad H_{y1}^- = -\frac{E_{x10}^-}{Z_1} e^{jk_1 z}$$

$$E_{x2}^+ = \underline{E_{x20}^+} e^{-jk_2(z-d)} \quad H_{y2}^+ = \frac{E_{x20}^+}{Z_2} e^{-jk_2(z-d)} \quad E_{x3}^+ = \underline{E_{x30}^+} e^{-jk_3(z-d)}$$

$$E_{x2}^- = \underline{E_{x20}^-} e^{jk_2(z-d)} \quad H_{y2}^- = -\frac{E_{x20}^-}{Z_2} e^{jk_2(z-d)} \quad H_{y3}^+ = \frac{E_{x30}^+}{Z_3} e^{-jk_3(z-d)}$$

2) 边界条件

$$z = d$$

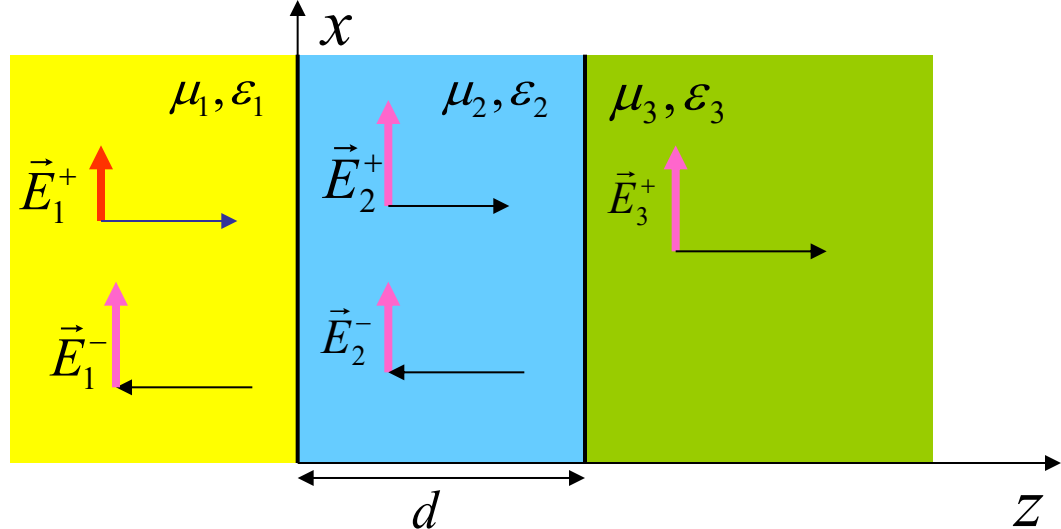
$$E_t: E_{x20}^+ + E_{x20}^- = E_{x30}^+$$

$$H_t: \frac{1}{Z_2} (E_{x20}^+ - E_{x20}^-) = \frac{E_{x30}^+}{Z_3}$$

$$z = 0$$

$$E_t: E_{x10}^+ + E_{x10}^- = E_{x20}^+ e^{jk_2 d} + E_{x20}^- e^{-jk_2 d}$$

$$H_t: \frac{1}{Z_1} (E_{x10}^+ - E_{x10}^-) = \frac{1}{Z_2} (E_{x20}^+ e^{jk_2 d} - E_{x20}^- e^{-jk_2 d})$$



$$\underline{E_{x20}^+} + \underline{E_{x20}^-} = \underline{E_{x30}^+}$$

$$\frac{1}{Z_2}(E_{x20}^+ - E_{x20}^-) = \frac{E_{x30}^+}{Z_3}$$

$$\underline{E_{x10}^+} + \underline{E_{x10}^-} = \underline{E_{x20}^+ e^{jk_2 d} + E_{x20}^- e^{-jk_2 d}} \quad E_{x2}(z=0)$$

$$\frac{1}{Z_1}(E_{x10}^+ - E_{x10}^-) = \frac{1}{\underline{Z_2}}(E_{x20}^+ e^{jk_2 d} - E_{x20}^- e^{-jk_2 d}) \quad H_{y2}(z=0)$$

$$\Rightarrow R_{23} = \frac{E_{x20}^-}{E_{x20}^+} = \frac{Z_3 - Z_2}{Z_3 + Z_2}$$

$$\Rightarrow T_{23} = \frac{E_{x30}^+}{E_{x20}^+} = \frac{2Z_3}{Z_3 + Z_2}$$

$$R_{12} = \frac{E_{x1}^-(z=0)}{E_{x1}^+(z=0)} = \frac{E_{x10}^-}{E_{x10}^+} \quad Z_{in2}(z=0) = \frac{E_{x2}(z=0)}{H_{y2}(z=0)}$$

$$\Rightarrow E_{x10}^- = R_{12} E_{x10}^+ \quad \Rightarrow Z_{in2}(z=0) = \frac{E_{x20}^+ e^{jk_2 d} + E_{x20}^- e^{-jk_2 d}}{E_{x20}^+ e^{jk_2 d} - E_{x20}^- e^{-jk_2 d}} Z_2$$

$$\frac{E_{x10}^+ + E_{x10}^-}{E_{x10}^+ - E_{x10}^-} Z_1 = Z_{in2}(0)$$

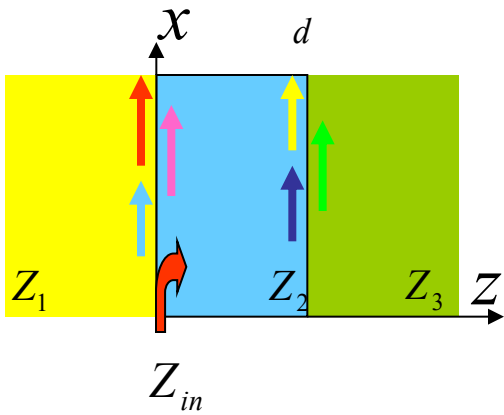
$$= Z_2 \frac{Z_3 + jZ_2 \operatorname{tg}(k_2 d)}{Z_2 + jZ_3 \operatorname{tg}(k_2 d)}$$

$$\frac{1 + \frac{E_{x10}^-}{E_{x10}^+}}{1 - \frac{E_{x10}^-}{E_{x10}^+}} Z_1 = \frac{1 + R_{12}}{1 - R_{12}} Z_1 = Z_{in2}(0) \quad \Rightarrow R_{12} = \frac{Z_{in2} - Z_1}{Z_{in2} + Z_1}$$

$$(E_{x10}^+ + R_{12} E_{x10}^+) = (E_{x20}^+ e^{jk_2 d} + R_{23} E_{x20}^+ e^{-jk_2 d})$$

$$\Rightarrow E_{x20}^+ = \frac{1 + R_{12}}{\underline{e^{jk_2 d} + R_{23} e^{-jk_2 d}}} E_{x10}^+$$

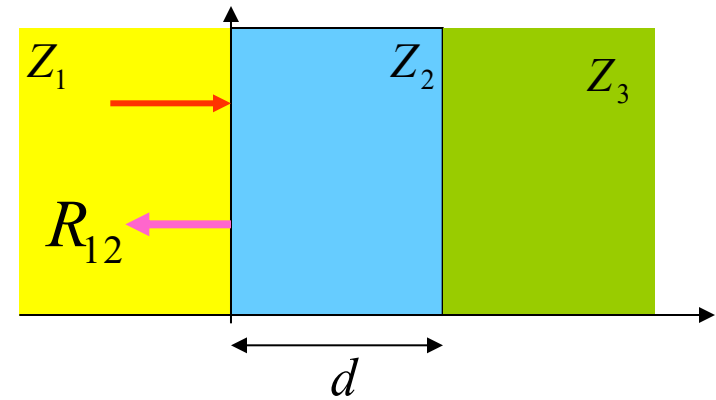
T_{12}



$$R_{12} = \frac{Z_{in2} - Z_1}{Z_{in2} + Z_1} \quad Z_{in2}(z=0) = Z_2 \frac{Z_3 + jZ_2 \operatorname{tg}(k_2 d)}{Z_2 + jZ_3 \operatorname{tg}(k_2 d)}$$

$$R_{12} = 0 \Rightarrow Z_{in2} = Z_1$$

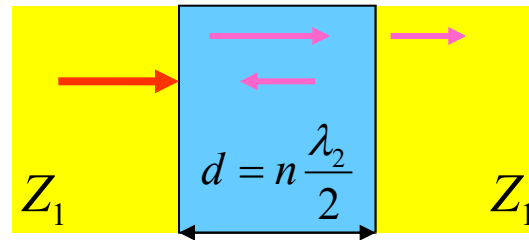
有两种情况:



$$(1) \quad k_2 d = n\pi \quad \operatorname{tg}(k_2 d) = 0$$

$$d = \frac{n\pi}{k_2} = n \frac{\lambda_2}{2}$$

$$Z_{in} = Z_3 = Z_1$$

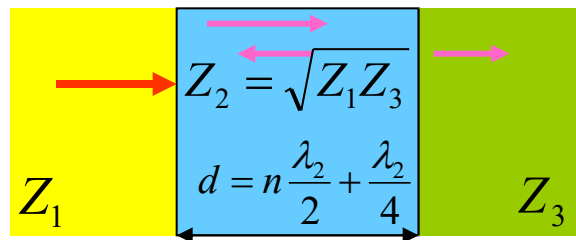


$$(2) \quad k_2 d = n\pi + \frac{\pi}{2} \quad \operatorname{tg}(k_2 d) = \infty$$

$$d = n \frac{\lambda_2}{2} + \frac{\lambda_2}{4}$$

$$Z_{in} = \frac{Z_2^2}{Z_3} = Z_1$$

$$Z_2 = \sqrt{Z_1 Z_3}$$



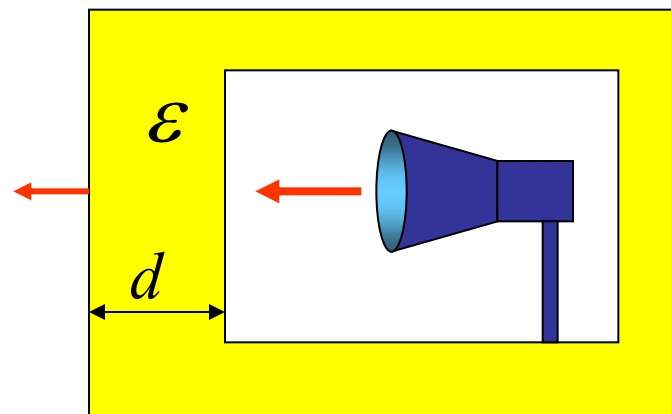
$$\mu_1 = \mu_2 = \mu_3 \quad \varepsilon_2 = \sqrt{\varepsilon_1 \varepsilon_3}$$

例1: $f = 4GHz$

天线罩 $\varepsilon_r = 2.25, \mu_r = 1$

求: $d = ?$

解: $Z_1 = Z_3 \quad d = \frac{\lambda_2}{2} = \frac{1}{2} \frac{c}{f} \frac{1}{\sqrt{\varepsilon_r}} = 2.5cm$



例2: $f = 6GHz \quad \mu_1 = \mu_2 = \mu_3 = \mu_0$

$\varepsilon_1 = \varepsilon_0 \quad \varepsilon_{r3} = 2.25$

求: $\varepsilon_{r2} = ?, d = ?$

解: $\varepsilon_{r2} = \sqrt{\varepsilon_{r1} \varepsilon_{r3}} = 1.5$

$$d = \frac{\lambda_2}{4} = \frac{1}{4} \frac{c}{f} \frac{1}{\sqrt{\varepsilon_{r2}}} = 0.0102m$$

